PV101 Day 4 Morning Quiz

PV wiring, polarity, protection, and small-system sequencing

CTS Cohort 6 - PV101 | Day 4 | Thursday, July 9, 2026

Topic: Small Solar System Build: PV Panel, Charge Controller, 12.8V LiFePO4 Battery, Wiring, and Inverter

Class setup: 6 students; 5 kits. 5 kits for 6 students: run 5 build stations. One station is a two-student team with Builder/Inspector roles that swap halfway. Solo builders still require a neighbor peer-check before energizing.

Name: _____ Date: _____

Directions: Answer in plain field language. Complete sentences are welcome; correct thinking matters more than fancy vocabulary.

1. 1. What does voltage describe in a circuit?

_____

_____

2. 2. What does current describe in a circuit?

_____

_____

3. 3. In a PV circuit, why do we verify polarity before connecting equipment?

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_____

4. 4. What usually happens to voltage when two matching modules are wired in series?

_____

_____

5. 5. What usually happens to current when two matching modules are wired in parallel?

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6. 6. Why should a fuse or breaker be close to the source of energy?

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7. 7. Why does a charge controller need correct battery connection and polarity?

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8. 8. Why does an inverter usually connect to the battery side instead of the small controller load terminals?

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9. 9. What should be checked before uncovering or connecting the PV panel?

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10. 10. Before energizing the small solar system, what should the group do?

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